

SCHOOL OF CHEMICAL & MATERIALS ENGINEERING
NATIONAL UNIVERSITY OF SCIENCES & TECHNOLOGY, ISLAMABAD
PSE-823 Advanced Process Dynamics and Control
Assignment 1 • Fall 2026

Full Name:	Reg #:
Topic: Open-loop dynamics, Python	Due Date: 30 Oct 2026, on Teams

Instructions

- Task 1 of each Part is a **derivation by hand**: submit clear photos or scans of your handwritten notes (one PDF); typed derivations are not accepted.
- Tasks 2–3 are coded in the starter notebook PSE823_Assignment1_Starter.ipynb (Colab). Enter your hand results in the first cell of each Part so the code can check them. Submit it as .ipynb or PDF.
- d = **last digit of your Reg #**; it sets the values below.
- Discuss freely, but derivations, code and answers must be your own.
- Notebook: *Restart and run all*, outputs visible, labelled axes with units; written answers must quote your own numbers.

Part	Content	Marks	Got
A	Tank (first order)	22	
B	Manometer (second order)	22	
C	Presentation	6	
Total		50	

Part A First-order system: liquid-level tank**22 marks**

Coughanowr Sec. 5.2, Prob. 5.2. A tank of area $A = 1.0 + 0.2d \text{ m}^2$ and height 6.0 m drains through $q_o = C\sqrt{h}$ (Eq. 5.30). At the operating point $q_s = 0.020 \text{ m}^3/\text{s}$ and $h_s = 2.0 \text{ m}$ (water).

A1 Derive by hand.**[7]**

- (a) From the mass balance (Eq. 5.31) **find** C , **linearize** about h_s and **obtain** $H/Q = R_1/(\tau s + 1)$ with numerical R_1 , τ and units. (b) **Write** the linear step response for $\Delta q = f q_s$. (c) For steps of +10%, +60%, –60% in q , **compute** the exact final level $(q/C)^2$ and the linear-model final level. (d) **Compute** the true time constant $2A\sqrt{h}/C$ at each new level.

A2 Code linear versus nonlinear.**[9]**

Enter your A1 values. **Write** `h_linear` and `h_nonlinear` (`solve_ivp`, Eq. 5.32); **plot** both for the three steps in three panels over $0 \leq t \leq 6\tau$, marking $h = 0$ and 6 m; **check** against `ct.step_response`. **Print** a table per step: hand final levels, max error (m), error as % of the change, apparent time constant (63.2% of the nonlinear change) and your hand τ from (d).

A3 Code and interpret.**[6]**

Plot $\tau(h_s)$ for $0.2 \leq h_s \leq 6 \text{ m}$ with the four levels marked, and **find** the largest step up and down (1% resolution) with linearization error $\leq 5\%$ of the change **[3]**. In at most six sentences with numbers, **explain**: for which steps the linear model is acceptable and why its error grows; what it predicts for –60% and why that is impossible; whether +60% overflows, and if the linear prediction is safe or unsafe for a high-level alarm **[3]**.

Part B Second-order system: U-tube manometer**22 marks**

Coughanowr Sec. 7.1, Prob. 7.5. Column length $L = 1.0 + 0.1d \text{ m}$, water at 20°C ($\rho = 998 \text{ kg/m}^3$, $\mu = 1.00 \text{ mPa s}$, $g = 9.81 \text{ m/s}^2$), laminar flow, $\beta = 4/3$, suddenly subjected to $\Delta P/\rho g = 0.20 \text{ m}$. Tube diameters $D = 1.0, 1.5, 2.5 \text{ mm}$.

B1 Derive by hand.**[7]**

- (a) From Eq. 7.3 **obtain** Eq. 7.4 and $\tau^2 Y'' + 2\zeta\tau Y' + Y = X$; **derive** τ and ζ (check the square roots yourself). (b) **Compute** τ , ζ for the three tubes and **classify** each. (c) For the underdamped tube **compute** overshoot, decay ratio and period (Eqs. 7.25–7.28). (d) **Find** the diameter giving $\zeta = 0.7$.

B2 Code three solutions and measurements.**[9]**

Enter your hand values. For each tube **compute** the step response by (i) your own analytic function covering all damping cases, (ii) `solve_ivp` (keep the velocity), (iii) `ct.step_response`; one plot, $0 \leq t \leq 8 \text{ s}$; **print** the largest differences. For the underdamped tube **measure** overshoot, decay ratio and period from the simulation and **compare** with your hand values; **report** the 5% response time of all three tubes.

B3 Code and interpret.**[6]**

For $0.05 \leq \zeta \leq 3$ **plot** overshoot and 5% response time/ τ and **report** the best ζ **[1]**; **simulate** your B1(d) design and **compute** the peak Reynolds number per tube (mean velocity $\frac{1}{2}|dh/dt|$) **[1]**; **split** the most overdamped tube into two lags (Eqs. 7.23–7.24, `ct.series`) and compare with one lag τ_1 **[1]**. In at most six sentences with numbers, **explain**: which tube you would fit and why; which lag makes the overdamped tube slow and why the other is invisible; whether laminar flow holds **[3]**.

Part C Presentation and reproducibility**6 marks**

The notebook runs top to bottom after *Restart and run all* **[3]**; figures have labelled axes, units and legends **[2]**; handwritten derivations are legible and code is commented with the symbols of the text **[1]**.